Jingxuan Fan

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EDUCATION

Harvard University Cambridge, MA

Ph.D. in Neurobiology May 2026

Dissertation: Sensory Motor Circuit Computation in *Drosophila* Goal-directed Navigation

M.S. in Applied Math March 2024

Massachusetts Institute of Technology Cambridge, MA

B.S. in Brain and Cognitive Sciences May 2020

Honors & Awards: Hans Lukas Teuber Award for Outstanding Academics, Walle J.H. Nauta Award for Outstanding Research

SELECTED WORK EXPERIENCE

Scale AI, MLRE *May. 2026-present*

- Enterprise deep research agent benchmarking and post-training

Amazon, Research Intern *June. 2025-Dec. 2025*

- Created a novel benchmark for evaluating LLMs task performance considering both model capability and personalized preference alignment; in the application case of personalized recommendation, developed a process reward metric to balance both recommendation adoption and evidence faithfulness and leveraged it to perform RL post-training (accepted at **NYRL**)

Meta, Research Intern *May 2024-Aug. 2024*

- Developed a novel image-based feature representation tailored to high density sEMG and used customized CV models for gesture decoding and input feature attributions
- Introduced manifold capacity as a theoretical metric for representation quality evaluation and multimodal SSL loss
- Implemented generative models to extract disentangled factors in sEMG for generalization and data augmentation

Axoft, Data Science Intern *Sept. 2023-Dec. 2023*

- Developed and maintained in-house software pipelines for fluorescence imaging processing and spike sorting
- Applied statistical and machine learning models for neural decoding from population spiking and LFP data

Merck Research Laboratories, Research Intern June 2018-August 2018

- Performed cell culture, immunoprecipitation and high throughput fluorescence imaging and quantification to characterize TDP-43 protein interaction networks and pathology in Amyotrophic Lateral Sclerosis (ALS) model

Harvard University Cambridge, MA

PhD Researcher, Dept. of Neurobiology *Sept. 2020-May 2026*

- Discovered heterogeneous locomotion- and directional goal-related tuning in the Drosophila sensory learning center; demonstrated emergence via dopamine-driven plasticity of sensory panorama learning; showed coordination of downstream motor programs through regulation of a coupled ring attractor (selected talk at **Janelia Grounding Cognition in Mechanistic Insight Conference**; journal manuscript in prep)
- Pretrained text-to-image diffusion models on designed text-image corpora and used controlled attention localization, ablation, and circuit discovery to study mechanisms for generating object color, shape, and spatial relationships; identified a general solution for correct spatial-relation generation and how it varies with text encoding and model scale (**CVPR 2026 Highlight**)
- Developed a synthetic benchmark for user–assistant information bias across diverse LLMs; showed how different post-training recipes shape this bias, and demonstrated bidirectional control through synthetic fine-tuning on real conversations (**NeurIPS Multi-Turn Interactions 2025, ACL 2026**)
- Studied continual knowledge injection in autoregressive and diffusion LMs, including the impact of paraphrase augmentation; showed diffusion LMs are more data-efficient and avoid the reversal curse, and developed masked training that closes the performance gap for autoregressive LLMs (**COLM Context Beyond the Window 2026**, **TMLR** submission)

Master's Researcher, Dept. of Applied Mathematics *Sept. 2020 - March 2024*

- Developed an entropy-penalized composition method for multi-attribute reward models and demonstrated improved results on reward model benchmarks (**AAAI 2026**)

- Developed a framework to generate large-scale synthetic rule pool and perform data-aware rule selection for scoring preference data in the safety domain; Demonstrated improved results on reward model benchmarks using preference data scoring with the rule adaptor (**ICML 2025**)
- Developed an automated method to generate a large-scale, domain-specific dataset of graduate-level applied mathematics problems and benchmarked frontier LLMs on this dataset (**NeurIPS MATH-AI 2024, ICLR 2025)**
- Developed next-token preference (NTP), an unsupervised, annotation-free framework that trains reward models from raw math web text, improving reward-model evaluation and downstream best-of-N selection, GRPO, and supervised RM training (**ICLR SPOT 2026, NeurIPS 2026)**

SELECTED PUBLICATIONS AND TALKS

Hoy, W., **Fan, J.,** Celik, N., Pan, X. (2026). Continuous Context Management. Submission to **ICLR 2027.**

Hoy, W., **Fan, J.,** Celik, N., Pan, X. (2026). Asynchronous Is Nearly Free for Evolution Strategies on Long-Horizon Agentic Tasks. Accepted at **NeurIPS 2026 Pre-to-Post Workshop.** Submission to **ARR.**

Chatrath, V.*, Zhu, B.*, **Fan, J.***, Pu, G., Tiwari, S. D., Dan, S., Young, R., Li, Y., Yao, Y., Shanker, A., Yang, M., Zhang, D. Y., He, Y., Liu, Y., Wang, C., Yin, Z., Xue, Y. (2026). READY or Not: Reliable Enterprise Agent Deployment. Submission to **ICLR 2027.**

Chatrath, V.*, Zhu, B.*, Pu, G., **Fan, J.**, Shanker, A., Ursekar, V., Sharma, A., Qin, J., Han, K., Tiwari, S. D., Dan, S., Kalmath, V., Li, Y., Zhang, D. Y., Wang, C., Doctor, Z., Yin, Z., Shah, N. H., Xue, Y. (2026). CliniCARE-Bench: Clinical Calibrated Audit of Medical Reasoning in EHR. Submission to Nature Communication.

Fan, J., Li, Y., Qi, Z., Zhang, D., Brantley, K., Kakade, S. M., Zhang, H. (2026). Scaling Reward Modeling without Human Supervision. **ICLR SPOT 2026**, **NeurIPS 2026**

Xu, P.*, Hahami, E.*, **Fan, J.*,** Xie, Z., Sompolinsky, H. (2025). Closing the Data-Efficiency Gap Between Autoregressive and Masked Diffusion LLMs. Submission to **TMLR**

Fan, J., Liu, H., Yuan, B. (2025). Measuring and optimizing evidence preference tradeoff in LLM personalized recommendation. **NYRL**

Xu, P.*, **Fan, J.*,** Xiong, Z., Hahami, E., Overwiening, J., Xie, Z. (2025). User-Assistant Bias in LLMs. **NeurIPS Multi-Turn Interactions 2025, ACL 2026**

Wang, B.*, **Fan, J.***, Xu, P.* (2025). Circuit Mechanisms for Spatial Relation Generation in Diffusion Transformers. **New England Mechanistic Interpretability (NEMI) workshop**, **CVPR 2026 highlight**

Fan, J., Wilson, R. (2025). Mapping a dynamic sensory panorama onto allocentric direction representations in goal-directed navigation. Selected talk at **Janelia Grounding Cognition in Mechanistic Insight Conference**

Mechanisms for balancing course stabilization and exploration. Talk at **Harvard Department of Neurobiology**.

Li, X., Chen X., **Fan, J.,** Gao, M., Jiang, H. (2025). Entropy-aware Attribute Composition of Multi-head Reward Models (https://arxiv.org/abs/2503.20995). **AAAI 2026**

Li, X.*, Gao, M.*, **Fan, J.**†, Zhang, Z.†, Li, W. (2025). Data-adaptive Safety Rules for Training Reward Models (https://arxiv.org/pdf/2501.15453). **ICLR BiAlign 2025, ICML 2025**

Fan, J., Martinson, S., Wang, E.Y., Hausknecht, K. (2024). HARDMath: A Benchmark Dataset for Challenging Problems in Applied Mathematics (https://arxiv.org/pdf/2410.09988). **NeurIPS 2024 MATH-AI workshop, ICLR 2025**

Kwon, J.-T., Ryu, C., Lee, H., Sheffield, A., **Fan, J.,** Cho, D. H., Bigler, S., Sullivan, H. A., Choe, H. K., Wickersham, I. R., Heiman, M., & Choi, G. B. (2021). An amygdala circuit that suppresses social engagement. **Nature**, 593(7857), 114–118.

Wei, H.*, Wiśniowska, A.*, **Fan, J.†,** Harvey, P.†, Li, Y., Wu, V., Hansen, E. C., Zhang, J., Kaul, M. G., Frey, A. M., Adam, G., Frenkel, A. I., Bawendi, M. G., & Jasanoff, A. (2021). Single-nanometer iron oxide nanoparticles as tissue-permeable MRI contrast agents. **Proceedings of the National Academy of Sciences**, 118(42).

Hsieh V., Okada S., Wei H., García-Álvarez I., Barandov A., Alvarado SR., Ohlendorf R., **Fan J.,** Ortega A., Jasanoff A. (2019). Neurotransmitter-responsive nanosensors for T2-weighted magnetic resonance imaging. **Journal of the American Chemical Society**, 141 (40), 15751-15